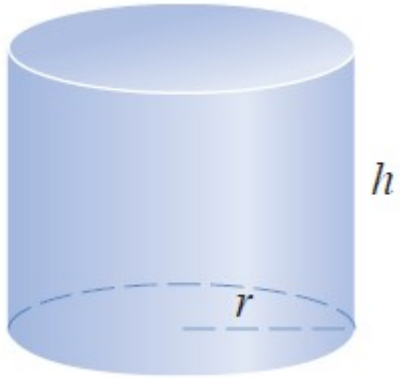
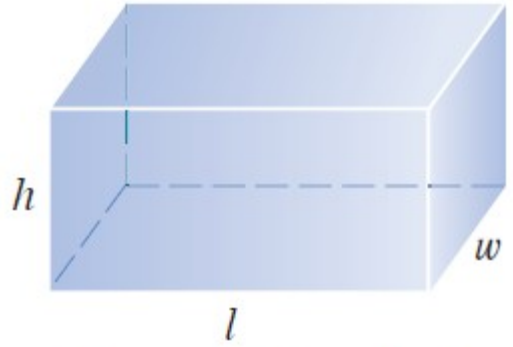
Chapter 7: Applications of Integration

Section 7.2: Volume: Disk Method
(**The Cross-Section Method**)

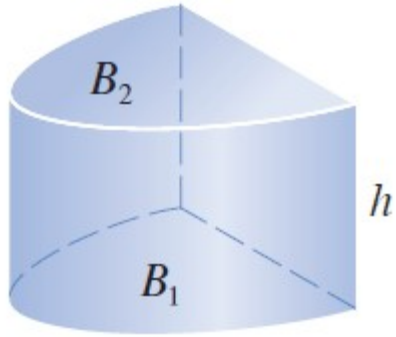
We start with a simple type of solid called a **cylinder** (or, more precisely, a *right cylind*



(b) Circular cylinder $V = \pi r^2 h$



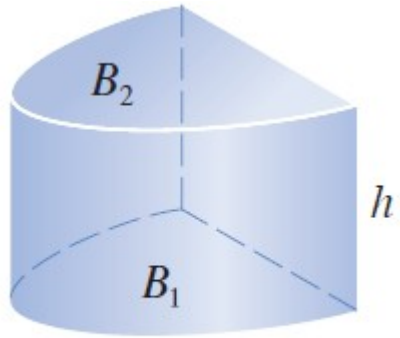
(c) Rectangular box $V = lwh$



(a) Cylinder

In fact, we define the volume of the cylinder as

$$V = Area\ of\ the\ base \times height = Ah$$

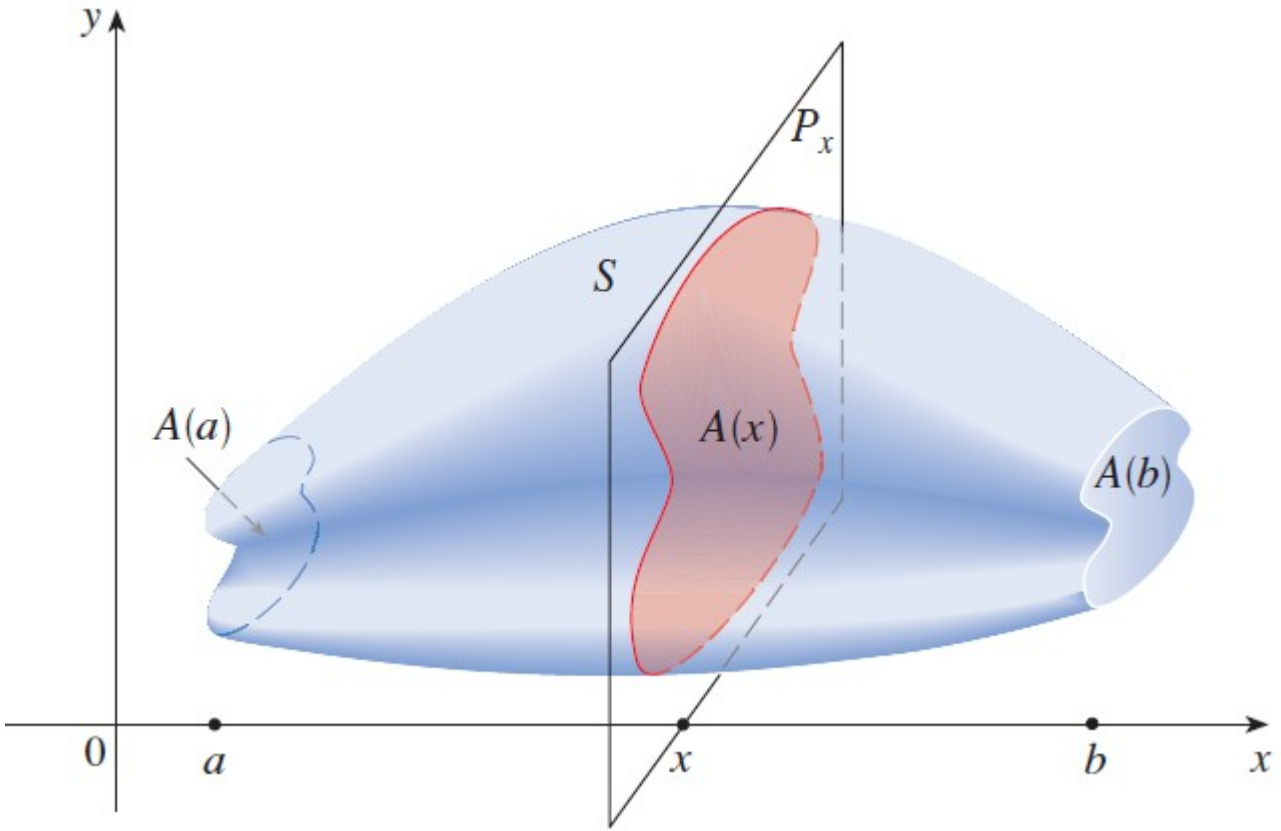


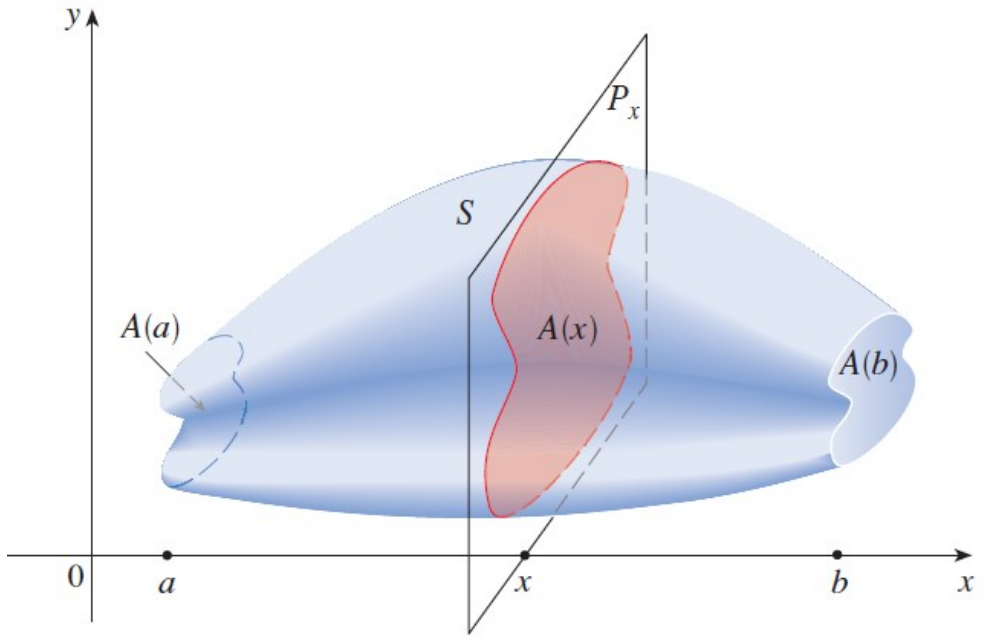
(a) Cylinder $V = Ah$

Q) How to find the volume of a solid S that is not a cylinder?

For a solid *S* that isn't a cylinder we first "cut" *S* into pieces and approximate each piece by a cylinder. We estimate the volume of *S* by adding the volumes of the cylinders. We arrive at the exact volume of *S* through a limiting process in which the number of pieces becomes large.

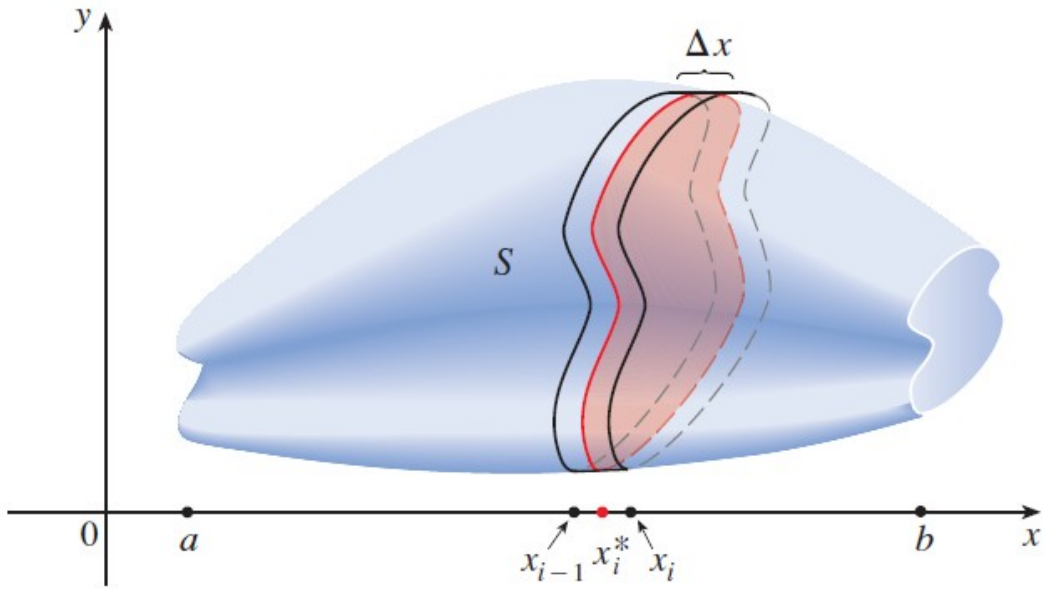
Definition: If a plane intersects a solid S, then the region common to the plane and S is called a **cross-section** of S.

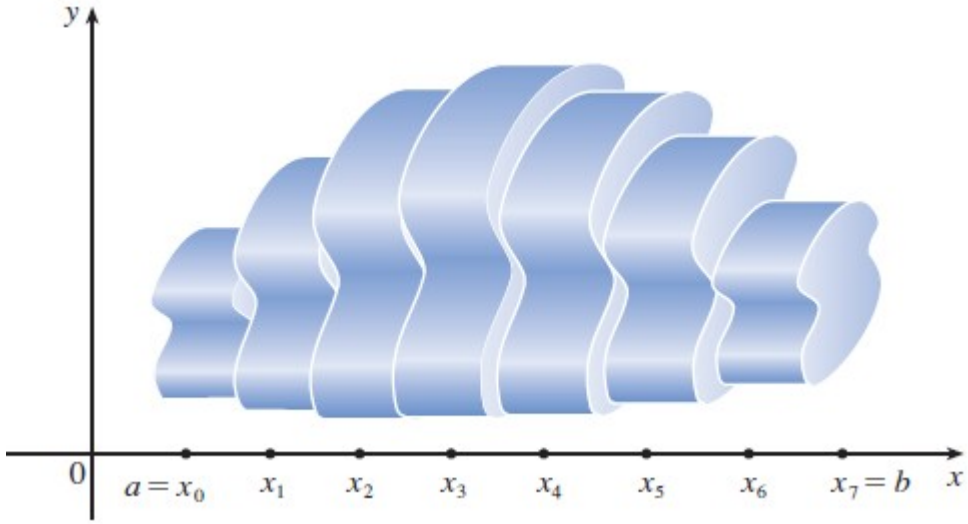




Let S be the solid bounded by planes that are perpendicular to the x-axis at a and b. Let $A(x)$ be the area of the cross-section for every $x \in [a, b]$.

To find the volume of S, we divide the interval $[a, b]$ into n subintervals which has the effect of sub-diving the solid into n **slabs** each of thickness $\Delta x = (b - a)/n$**.**





The volume of a typical slab can be approximated by ΔV where

$$\Delta \text{V} = \text{Area of the cross-section} \times \text{Thickness} \quad = A(x_i^*)\Delta x$$

Where we are assuming that A is a continuous function.

Then the required volume of the solid S is given by

$$V = \lim_{n \to \infty} \sum_{i=1}^{n} A(x_i^*)\,\Delta x = \int_a^b A(x)\,dx$$

Definition of Volume Let S be a solid that lies between $x = a$ and $x = b$. If the cross-sectional area of S in the plane P_x, through x and perpendicular to the x-axis, is $A(x)$, where A is a continuous function, then the **volume** of S is

$$V = \lim_{n \to \infty} \sum_{i=1}^{n} A(x_i^*)\,\Delta x = \int_a^b A(x)\,dx$$

When we use the volume formula $V = \int_a^b A(x)\,dx$, it is important to remember that $A(x)$ is the area of a moving cross-section obtained by slicing through x perpendicular to the x-axis.

Notice that, for a cylinder, the cross-sectional area is constant: $A(x) = A$ for all x. So our definition of volume gives $V = \int_a^b A\,dx = A(b - a)$; this agrees with the formula $V = Ah$.

EXAMPLE **Try this website to get a visualization of the solid** https://www.geogebra.org/m/yDpSgNG6

The base of a solid S is bounded by $y = \sqrt{1 - x}$ and the x-axis from $x = 0$ to $x = 1$.
Find the volume of the solid S if the cross-sections perpendicular to the x-axis are **squares**.

SOLUTION

$$V = \frac{1}{2}$$

EXAMPLE **Try this website to get a visualization of the solid** https://www.geogebra.org/m/CdZ7btrB

The base of a solid S is bounded by $y = \sqrt{1 - x}$ and the x-axis from $x = 0$ to $x = 1$.
Find the volume of the solid S if the cross-sections perpendicular to the ***y*-axis** are **equilateral triangles**.

Area of equilateral triangle

$$\text{Area} = \frac{\sqrt{3}}{4}(\text{base})^2$$

SOLUTION

$$V = \frac{2\sqrt{3}}{15}$$

EXAMPLE **Try this website to get a visualization of the solid** https://www.geogebra.org/m/CdZ7btrB

The base of a solid S is bounded by $y = \sqrt{1 - x}$ and the x-axis from $x = 0$ to $x = 1$.
Find the volume of the solid S if the cross-sections perpendicular to the ***y*-axis** are **semicircles**.

SOLUTION

$$V = \frac{\pi}{15}$$

EXAMPLE 7 An Application to Geometry

Prove that the volume of a pyramid with a square base is

$$V = \frac{1}{3}hB$$

where h is the height of the pyramid and B is the area of the base.

Do it or read the solution form the textbook

Volumes of Solids of Revolution :The Disk Method

If we revolve a region about a line, we obtain a **solid of revolution**.

If a plate region rotates about a given axis, which we call it axis of rotation, then the cross-sections to the resulting solid are **disks**. Hence the area of a typical cross-section is

$$A = \pi\,(radius)^2$$

Try this website to get a visualization these type of solids https://www.geogebra.org/classic/Hw7bnB2q

EXAMPLE **Try this website to get a visualization these type of solids** https://www.geogebra.org/classic/Hw7bnB2q

Find the volume of the solid obtained by rotating the region bounded by $y = \sqrt{x},\ 0 \le x \le 4$, about the x-axis.

SOLUTION

$V = 8\pi$

EXAMPLE **Try this website to get a visualization these type of solids** https://www.geogebra.org/classic/Hw7bnB2q

The region bounded by y-axis, the graph of $y = e^x$ and the line $y = 3$ is revolved about the line $y = 3$.
Find the volume of the resulting solid.

SOLUTION

$$V = \pi(9\ln 3 - 8)$$

EXAMPLE **Try this website to get a visualization these type of solids** https://www.geogebra.org/classic/Hw7bnB2q

Set up the integral that can be used to evaluate the volume of the solid generated by revolving the region bounded by y-axis, the graph of $y = e^x$ and the line $y = 3$ about the y-axis.

SOLUTION

$$V = \int_1^3 \pi \, (\ln y)^2 \, dy$$

EXAMPLE **Try this website to get a visualization these type of solids** https://www.geogebra.org/classic/Hw7bnB2q

Find the volume of the solid generated by revolving the region bounded by the curves $x = 4y^2 - y^3$ and $x = 0$ about the y-axis.

SOLUTION

$$V = \frac{16384\pi}{105}$$

Volumes of Solids of Revolution :The Washer Method

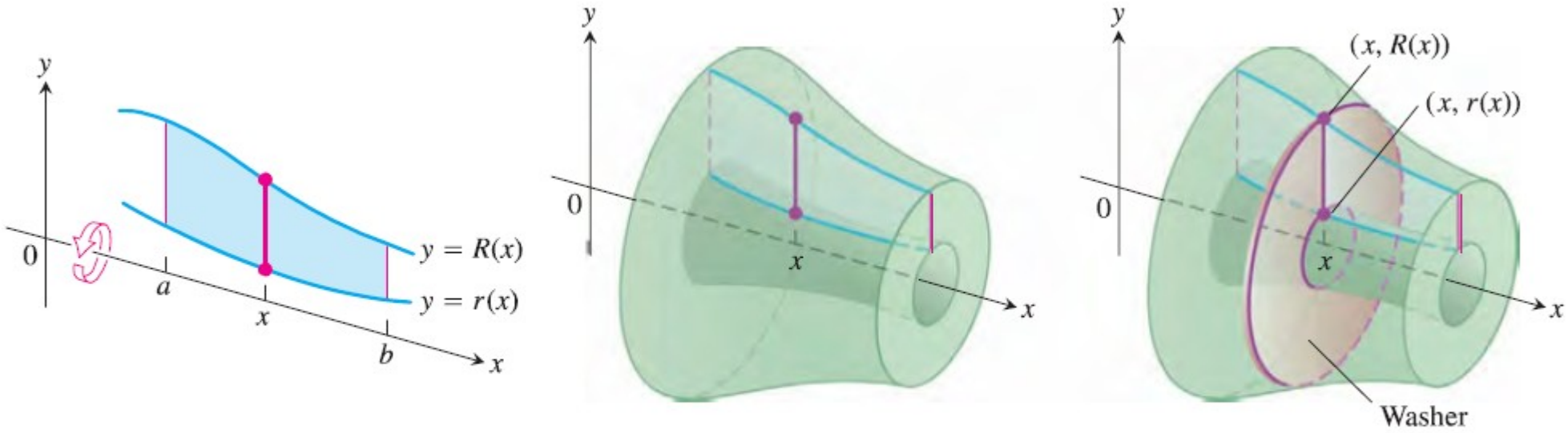
If the region we revolve to generate a solid does not border on or cross the axis of revolution, the solid has a hole in it (see the figure below). The cross-sections perpendicular to the axis of revolution are *washers* (the purplish circular surface in the figure) instead of disks. The dimensions of a typical washer are

Outer radius: $R(x)$

Inner radius: $r(x)$

The washer's area is

$$A(x) = \pi[R(x)]^2 - \pi[r(x)]^2 = \pi([R(x)]^2 - [r(x)]^2).$$



The cross-sections of the solid of revolution generated here are washers, not disks, so the integral $\int_a^b A(x)\,dx$ leads to a slightly different formula.

EXAMPLE **Try this website to get a visualization these type of solids** https://www.geogebra.org/classic/Hw7bnB2q

Find the volume of the solid generated by revolving the region bounded by the curves $y = x^3$ and $y = x, x \geq 0$ about the x-axis.

SOLUTION

$$V = \frac{4\pi}{21}$$

EXAMPLE **Try this website to get a visualization these type of solids** https://www.geogebra.org/classic/Hw7bnB2q

Find the volume of the solid generated by revolving the region bounded by the curves $y = \sin x$ and $y = \cos x$, $0 \le x \le \pi/4$ about $y = -1$.

SOLUTION

$$V = \pi\left(2\sqrt{2} - \frac{3}{2}\right)$$

EXAMPLE **Try this website to get a visualization these type of solids** https://www.geogebra.org/classic/Hw7bnB2q

Set up the integral (but do not evaluate) that can be used to find the volume of the solid generated by rotating the region enclosed by the curves $y = x^2$ and $y^2 = 2x$ about the y-axis.

SOLUTION

$$V = \pi \int_0^{\sqrt[3]{4}} \left(y - \frac{1}{4} y^4 \right) \, dy$$

EXAMPLE **Try this website to get a visualization these type of solids** https://www.geogebra.org/classic/Hw7bnB2q

Set up the integral (but do not evaluate) that can be sued to find the volume of the solid generated by rotating the region enclosed by the curves $y = x^2$ and $y^2 = 2x$ about $x = 4$.

SOLUTION

$$V = \pi \int_0^{\sqrt[3]{4}} \left(\left(4 - \frac{y^2}{2}\right)^2 - (4 - \sqrt{y})^2 \right) \, dy$$

Solve exercises of section 7.2.

Start with the suggested problems mentioned in the syllabus.